

Alexis Ruiz

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SUMMARY

Mechanical engineering graduate student seeking internship and full-time experience with hands-on experience in mechanical design, simulation, and prototyping. Skilled in applying analytical and computational tools to model and analyze complex systems.

EDUCATION

Master of Science Mechanical Engineering, Dec 2026

University of Wisconsin-Madison

Ph.D. Mechanical Engineering (all but dissertation), January 2023 – Present

Master of Science Mechanical Engineering, December 2022

California State University, Los Angeles

Dean's List (4 semesters)

Coursework: Kinematics and Dynamics of Machine Systems, Machine Design, Fatigue and Failure, Scientific Computing and ML for Engineering, Artificial Neural Networks, Data Driven Engineering Design, Advanced Computational Dynamics, Precision Measurements

ENGINEERING EXPERIENCE

Simulation Based Engineering Lab, Department of Mechanical Engineering, UW-Madison, WI

Research Assistant, *January 2023 – Present*

- Developed and validated physics-based simulation models of a fully electric, 4-wheeled independent-drive vehicle, incorporating terrain models (SCM/CRM) for performance calibration against real-world test data.
- Improved the mechanical design of a 3D-printed vehicle platform by identifying structural flaws, redesigning components, and producing an optimized complex wheel geometry for sandy terrain performance.
- Designed custom sensor and component mounts for modified off-the-shelf RC vehicles, ensuring robust mechanical and electrical integration for autonomous testing applications.
- Designed a wireless, TCP-based data acquisition system that collects data from multiple devices simultaneously, streamlining hardware testing.
- Supervised hardware assets and lab hardware projects.

Robotics Laboratory, Department of Mechanical Engineering, CSULA, CA

Research Assistant, *January 2020 – December 2022*

- Improved power generation in vertical-axis wind turbines (VAWTs) by implementing active blade pitching controls via reinforcement learning algorithms to adapt to dynamic wind conditions in a simulated environment, culminating with a publication in the journal, *Energy*.
- Worked hands-on to build a hardware prototype platform of a VAWT incorporating electrical systems, a custom manufactured frame, and 3D printed turbine blades.
- Conducted wind tunnel test of a VAWT prototype with active blade pitch controls and obtained preliminary results, leading to thrust in parameter estimation and calibration.
- Conducted research on bipedal robot dynamics, controls, and machine learning trained gaits.
- Utilized SolidWorks and MATLAB for physical model simulation and validation.

SKILLS & TECHNICAL COMPETENCIES

Programming: C, C++, MATLAB, Python, CUDA

Embedded Systems: Arduino, Microcontroller Programming, Sensor Integration, Serial Communications, ControlLogix 5575 (Programming and Integration)

Software Proficiency: SolidWorks, AutoCAD, Patran, LTspice, LabVIEW, Simulink, Studio 5000, Git, Docker

Technical Skills: FEA, Linux, ROS, Power Tools, Manual Machining

Certifications: EIT – NCEES No. 159362

Global Languages: Spanish

EXPERIENCE

Teaching Assistant (Dynamic Systems), *August 2025 – Present*

Department of Mechanical Engineering, UW–Madison, WI

- Independently lead discussion sessions, clarifying lecture material and resolving student questions.
- Acted as primary liaison between students and professor, facilitating effective communication and feedback.

Teaching Assistant (Senior Capstone Project and Robotics), *February 2020 – December 2022*

Department of Mechanical Engineering, CSULA, CA

- Supported the senior capstone program by addressing diverse project needs in hardware and electrical systems, including model building, simulation and deployment of prototypes.
- Collaborated with the senior capstone director to guide student groups in small UAV modeling and control using MATLAB and Pixhawk PX4 control units.
- Developed and presented a semester-long project in robotics, guiding students in integrating software and hardware applications.

OUTREACH

Student Outreach Mentor, ProCSI-C, UW-Madison

- Hosted visiting college students and introduced them to campus academic life.
- Guided visiting students through manufacturing shops, makerspaces, research labs, and available resources present in the department.
- Explained ongoing research projects present in the lab and facilitated discussions with faculty and lab members.

PUBLICATIONS

Ruiz, A., Zhang, H., Unjhwala, H.M., Zhou, Z., Radu, S., Negrut, D. Simultaneous Rover and Soil Model Calibration via Bayesian Optimization (*In preparation*).

Zhang, R., Tagliaferro, B., Warzecha, M., **Ruiz, A.**, Negrut, D. Data-driven rover wheel design using Gaussian Processes surrogate modeling (*Submitted*).

Zhang, H., Unjhwala, H.M., Young, A., **Ruiz, A.**, Bakke, L, Radu, S., Negrut, D. Simulation-Based Development of Path Following Control Policies for Autonomous Ground Robots: A Case Study (*Submitted*).

Zhou, Z., Unjhwala, H.M., Kamaraj, A., Schwarz, C., **Ruiz, A.**, Radu, S., Ahn, S., Lee, J., Negrut, D. A Framework for Large-Scale Heterogenous Traffic Simulation with Human In the Loop (*Submitted*).

Shen H., **Ruiz A.**, Li N. Fast online reinforcement learning control of small lift-driven vertical axis wind turbines with an active programmable four bar linkage mechanism. *Energy*. Vol 262, p. 125350, 2023.

Ruiz A. Intelligent Control of Vertical Axis Wind Turbines for High Efficiency Energy Generation. Master's Thesis, California State University, Los Angeles, 2022.

PRESENTATION AND POSTERS

Ruiz, A., Negrut, D. Cobra – A Hardware Validation Platform with Simulation Studies. Poster, Machine-Ground Interaction Consortium, Madison, WI, United States, 2025.

Ruiz, A., Ashokkumar, A. Overview of robotics hardware assets used in sim2real research in the Simulation-Based Engineering Lab. Oral presentation, Machine-Ground Interaction Consortium, Madison, WI, United States, 2023.

Ruiz, A. Bipedal Robot Control using Reinforcement Learning. Oral presentation, Student Symposium on Research, Scholarship, and Creative Activities, Los Angeles, CA, United States, 2021.